

torch.kron#

<https://pytorch.org/docs/stable/generated/torch.kron.html#torch.kron>

Description:

Computes the Kronecker product, denoted by $\otimes$, of input and other. If input is a $(a_0 \times a_1 \times \dots \times a_n)$ tensor and other is a $(b_0 \times b_1 \times \dots \times b_n)$ tensor, the result will be a $(a_0 \times b_0 \times a_1 \times b_1 \times \dots \times a_n \times b_n)$ tensor with the following entries: where $k_t = i_t \times b_t + j_t$ for $0 \leq t \leq n$. If one tensor has fewer dimensions than the other it is unsqueezed until it has the same number of dimensions. Supports real-valued and complex-valued inputs.

Function Signature

`torch.kron(input, other, *, out=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – The output tensor. Ignored if None. Default: None

Code Examples

```
>>> mat1 = torch.eye(2)
>>> mat2 = torch.ones(2, 2)
>>> torch.kron(mat1, mat2)
tensor([[1., 1., 0., 0.],
        [1., 1., 0., 0.],
        [0., 0., 1., 1.],
        [0., 0., 1., 1.]])

>>> mat1 = torch.eye(2)
>>> mat2 = torch.arange(1, 5).reshape(2, 2)
>>> torch.kron(mat1, mat2)
tensor([[1., 2., 0., 0.],
        [3., 4., 0., 0.],
        [0., 0., 1., 2.],
        [0., 0., 3., 4.]])
```

Notes & Warnings

NOTE

Note This function generalizes the typical definition of the Kronecker product for two matrices to two tensors, as described above. When input is a $(m \times n)$ matrix and other is a $(p \times q)$ matrix, the result will be a $(p * m \times q * n)$ block matrix:

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}$$

where input is A and other is B .

Detailed Documentation

Documentation

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If input is a $(a_0 \times a_1 \times \dots \times a_n)$ tensor and other is a $(b_0 \times b_1 \times \dots \times b_n)$ tensor, the result will be a $(a_0 * b_0 \times a_1 * b_1 \times \dots \times a_n * b_n)$ tensor with the following entries:

where $k_t = i_t * b_t + j_t$ for $0 \leq t \leq n$. If one tensor has fewer dimensions than the other it is unsqueezed until it has the same number of dimensions.

Supports real-valued and complex-valued inputs.

This function generalizes the typical definition of the Kronecker product for two matrices to two tensors, as described above. When input is a $(m \times n)$ matrix and other is a $(p \times q)$ matrix, the result will be a $(p * m \times q * n)$ block matrix:

where input is A and other is B .

out (Tensor, optional) – The output tensor. Ignored if None. Default: None